AP CALCULUS AB

Unit 1 • Lesson 1.7 notes

Lesson 1.7 • Selecting Procedures for Determining Limits

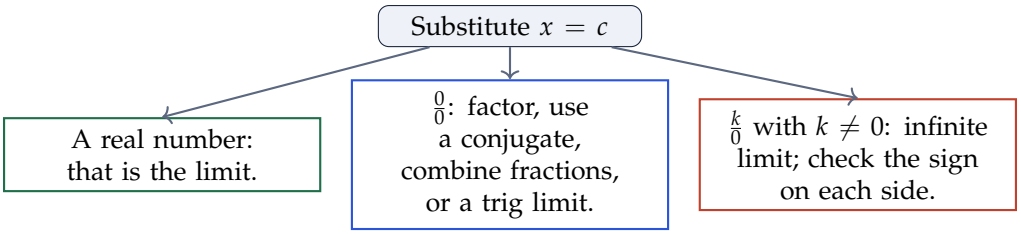
Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Choose an efficient method for a limit based on what substitution gives.
- Handle $\frac{k}{0}$ forms, absolute values and piecewise functions.

Vocabulary. direct substitution, indeterminate, infinite limit, one-sided analysis.

1. A decision procedure



For absolute values and piecewise functions, compute each one-sided limit separately.

Example 1 • $\frac{k}{0}$

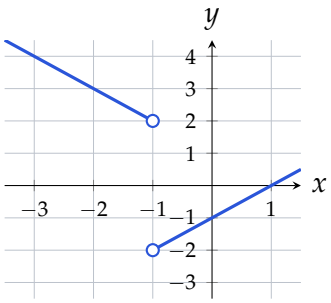
$\lim_{x\to 2} \dfrac{x+3}{x-2}$: substitution gives $\frac{5}{0}$.

Solution. For $x < 2$ the denominator is negative: $\lim_{x\to 2^-} = -\infty$. For $x > 2$: $\lim_{x\to 2^+} = +\infty$. The two-sided limit does not exist (not even as $\pm\infty$).

Example 2 • Absolute value

$\lim_{x\to -1} \dfrac{x^2-1}{|x+1|}$.

Solution. For $x < -1$: $|x+1| = -(x+1)$, so the quotient is $-(x-1) \to 2$. For $x > -1$: it is $x - 1 \to -2$. The limit does not exist.



Common mistakes

- Treating $\frac{5}{0}$ like $\frac{0}{0}$ and trying to factor.
- Writing $|x+1| = x+1$ for all x.

Quick check

1. $\lim_{x \to 0} \dfrac{x^2 - 3x}{x}$ 2. $\lim_{x \to 3^-} \dfrac{1}{x - 3}$ 3. $\lim_{x \to 2} \dfrac{x^2 + 1}{x + 3}$

Answers. 1. -3. 2. $-\infty$. 3. 1.